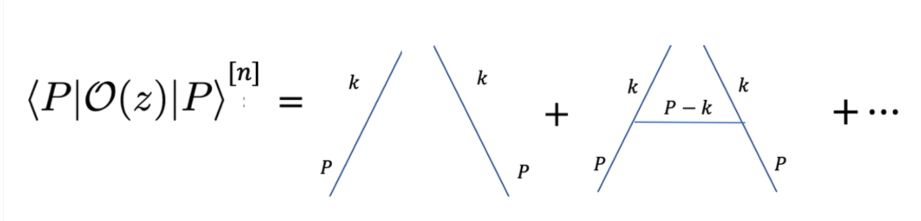


Figure 1: Zeroth and first order Feynman diagrams

The zeroth order Feynman diagram is $\Phi\big(k^2,\, k\!\cdot\!P,\, m^2\big)^{[0]} = \delta(P-k)$

The first order Feynman diagram is. $\Phi\big(k^2,\, k\!\cdot\!P,\, m^2\big)^{[1]} = \frac{\lambda^2}{(2\pi)^4}\,\frac{1}{(k^2-m^2)^2}\,(2\pi)\,\delta_+\big((P-k)^2-m^2\big)$.

1 Φ^3-theory in $4D$ (Super-renormalizable)

The k_T integrals are finite: no UV divergence. There is no operator so there is no distinction between f_1, g_1, h_1

1.1 PDFs

$$f(x) \;=\; xP^+ \int \mathrm{d}^2k_\perp\, dk^-\, \Phi\big(x, k_\perp^2, k^-\, m^2\big)\Big|_{x=k^+/P^+} \tag{1}$$

Now we plug in the model's correlator

$$\begin{aligned} f(x) &= xP^+ \int \frac{dk^- d^2k_T}{(2\pi)^4}\ \ (ig)\ \frac{i}{k^2-m^2}\,\frac{i}{k^2-m^2}(ig)\ \ 2\pi\ \ \delta_+\Big((P-k)^2-\lambda^2\Big)\Big|_{x=k^+/P^+} \\ &= g^2 xP^+ \int \frac{d^2k_T}{(2\pi)^3}\ \ \frac{dk^-}{(2xP^+k^- - k_T^2 - m^2)^2}\,\frac{1}{2P^+(1-x)}\delta\Big(k^- - \frac{M^2}{2P^+} + \frac{k_T^2+\lambda^2}{2P^+(1-x)}\Big)\Big|_{x=k^+/P^+} \end{aligned} \tag{2}$$

We take the k^- integral using the delta function

$$f(x) \;=\; \frac{g^2}{4\pi}\int \frac{d^2k_T}{(2\pi)^2}\ \frac{x(1-x)}{(k_T^2+u)^2} \qquad \text{with} \quad u = -x(1-x)M^2 + (1-x)m^2 + x\lambda^2 \tag{3}$$

The k_T integral is finite; we do not need any regularization

$$f(x) \;=\; \frac{g^2}{16\pi^2}\frac{x(1-x)}{\Big(-x(1-x)M^2+(1-x)m^2+x\lambda^2\Big)} \tag{4}$$

For $M = m = \lambda$

$$f(x) = \frac{g^2}{16\ m^2\ \pi^2} \frac{x(1-x)}{(x^2-x+1)} \tag{5}$$

1.2 pPDFs

Now there is a FT when taking the transverse integral. (The point spitting is still in the +-direction)

$$f(x, z_\perp) = xP^+ \int \mathrm{d}^2 k_\perp\, e^{ik_\perp . z_\perp}\ dk^-\ \Phi\big(k^-, k_\perp^2, x, m^2\big). \tag{6}$$

The k^- integral is the same as before, so we can start with

$$f(x, z_\perp) = \frac{g^2}{4\pi} x(1-x) \int \frac{d^2 k_T}{(2\pi)^2} \frac{e^{ik_T . z_T}}{(k_\perp^2 + u)^2} \tag{7}$$

Using the standard d-dimensional FT

$$\int \frac{d^D k}{(2\pi)^D} \frac{e^{ik\cdot z}}{(k^2+u)^\nu} = \frac{1}{(2\pi)^{D/2}} \frac{1}{2^{\nu-1}\Gamma(\nu)} \left(\frac{|z|}{\sqrt{u}}\right)^{\nu - D/2} K_{\nu-D/2}\Big(\sqrt{u\ z^2}\Big)\,. \tag{8}$$

$$f(x, z_\perp) = \frac{g^2}{16\pi^2} x(1-x) \frac{|z_T|}{\sqrt{u}} K_1\Big(\sqrt{z_T^2 u}\Big) \tag{9}$$

Using the small argument expansion of the Bessel function is

$$K_1(y) = \frac{1}{y} + \frac{y}{2}\left(\ln\frac{y}{2} + \gamma_E - \frac{1}{2}\right) + O\big(y^3 \ln y\big)\,.$$

$$\begin{aligned} f(x, z_\perp)_{Truncated} &= \frac{g^2}{16\pi^2} x(1-x) \frac{|z_T|}{\sqrt{u}} \frac{1}{\sqrt{z_T^2 u}} \\ &= \frac{g^2}{16\pi^2} \frac{x(1-x)}{\Big[-(x(1-x)M^2 + (1-x)m^2 + x\lambda^2\Big]} \end{aligned} \tag{10}$$

For $m = M = \lambda$

$$f(x, z_\perp)_{Truncated} = \frac{g^2}{16 m^2 \pi^2} \frac{x(1-x)}{(x^2 - x + 1)} \tag{11}$$

1.3 Zeroth order matching in ϕ^3-4D

$$f(x) = \frac{g^2}{16\ m^2\ \pi^2} \frac{x(1-x)}{(x^2 - x + 1)} \tag{12}$$

$$f(x, z_\perp)_{Truncated} = \frac{g^2}{16 m^2 \pi^2} \frac{x(1-x)}{(x^2 - x + 1)} \tag{13}$$

$$f(x, z_\perp)_{Truncated} - f(x) = 0 \tag{14}$$

2 Φ^3-theory in $6D$

2.1 PDF

All is the same as previous until taking the transverse integrals which is now divergent. We'll use dim reg and MS bar to calculate it.

$$f(x, \mu) = \frac{g^2}{4\pi} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} k_T}{(2\pi)^{4-2\epsilon}} \frac{x(1-x)}{(k_T^2 + u)^2} \qquad \text{with} \quad u = -x(1-x)M^2 + (1-x)m^2 + x\lambda^2 \tag{15}$$

We take the transverse integrals using formulae below for rotationally symmetric Euclidean integral in d dimesions:

$$\int d^d k f(k^2) = \frac{\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty dk^2 (k^2)^{d/2-1} f(k^2) \tag{16}$$

$$\int_0^\infty dx \frac{x^{\alpha-1}}{(x+A)^\beta} = A^{\alpha-\beta} \frac{\Gamma(\alpha)\Gamma(\beta-\alpha)}{\Gamma(\beta)} \tag{17}$$

Using these the transverse integral is

$$\int \frac{d^{4-2\epsilon}k_T}{(k_\perp^2+u)^2} = \pi^{2-\epsilon}\frac{\Gamma(\epsilon)}{u^\epsilon} \tag{18}$$

Using this result the PDF becomes

$$f(x,\mu) \quad = \quad \frac{g^2}{4\pi}x(1-x)\mu^{2\epsilon}\frac{\pi^{2-\epsilon}}{(2\pi)^{4-2\epsilon}}\frac{\Gamma(\epsilon)}{u^\epsilon} \tag{19}$$

Expanding in ϵ

$$f(x,\mu) = \frac{g^2}{(4\pi)^3}\ x(1-x)\left(\left(\frac{1}{\epsilon}-\gamma_E+\ln 4\pi\right)+\ln\frac{\mu^2}{u}\right)+\mathcal{O}(\epsilon) \tag{20}$$

Taking $\epsilon \to 0$

$$f(x,\mu) = \frac{g^2}{(4\pi)^3}\ x(1-x)\left(\mathcal{D}_{UV}+\ln\frac{\mu^2}{u}\right) \tag{21}$$

where $\mathcal{D}_{UV} = \left(\frac{1}{\epsilon}-\gamma_E+\ln 4\pi\right)$ is the UV divergence.

2.2 pPDFs

The FT regularizes the transverse integral

$$f(x,z_\perp) \quad = \quad \frac{g^2}{4\pi}x(1-x)\int\frac{d^4k_T}{(2\pi)^4}\ \ \frac{e^{ik_T\cdot z_T}}{(k_T^2+u)^2} \qquad \text{with} \quad u=-x(1-x)M^2+(1-x)m^2+x\lambda^2 \tag{22}$$

$$\begin{aligned} f(x,z_\perp) \quad &= \quad \frac{g^2}{4\pi}x(1-x)\ \ \frac{1}{8\pi^2}K_0(\sqrt{uz^2}) \\ &= \quad \frac{g^2}{32\pi^3}x(1-x)\ \ K_0(\sqrt{uz^2}) \end{aligned} \tag{23}$$

Small-argument expansion of the modified Bessel function $K_0(y)$

$$K_0(y) = -\gamma_E - \ln(\frac{y}{2}) + O(y^2), \qquad y>0. \tag{24}$$

$$
\begin{aligned}
f(x,z_\perp)_{Truncated} &= \frac{g^2}{32\pi^3}x(1-x)\ \Big[-\gamma_E - \ln\big(\frac{|z|\sqrt{u}}{2}\ \big)\Big] \\
&= \frac{g^2}{64\pi^3}x(1-x)\ \Big[-2\gamma_E + \ln 4 - \ln\big(z^2 u\big)\ \Big]
\end{aligned} \tag{25}
$$

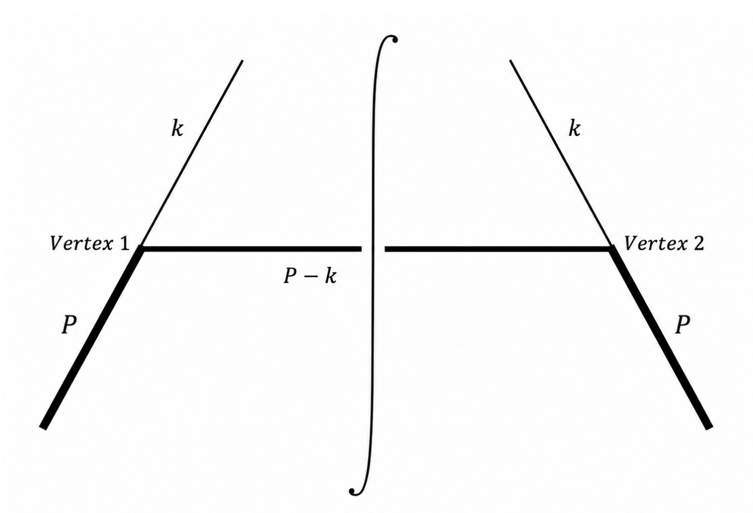
2.3 Zeroth order matching in ϕ^3-6D

$$
f(x,\mu) = \frac{g^2}{(4\pi)^3}\ x(1-x)\ln\frac{\mu^2}{u} \qquad with \qquad D_{UV}=0 \tag{26}
$$

$$
f(x,z_\perp)_{Truncated} = \frac{g^2}{64\pi^3}x(1-x)\ \Big[-2\gamma_E + \ln 4 - \ln\big(z^2 u\big)\ \Big] \tag{27}
$$

$$
\begin{aligned}
f(x,z_\perp)_{Truncated} - f(x) &= \frac{g^2}{64\pi^3}x(1-x)\ \Big[-2\gamma_E + \ln 4 - \ln\big(z^2 u\big) - \ln\frac{\mu^2}{u}\Big] \\
&= \frac{g^2}{64\pi^3}x(1-x)\ \Big[-2\gamma_E + \ln 4 - \ln\big(\mu^2 z^2\big)\Big]
\end{aligned} \tag{28}
$$

3 Scalar Diquark Model: f_1



3.1 PDF

The TMD is

$$\begin{aligned} f(x,k_\perp^2) &= \int \frac{dk^+dk^-}{(2\pi)^2} P^+\delta\Big(k^+-xP^+\Big)\,\bar{u}(P,S)\,(ig_s)\,\frac{i(\not{k}+m)}{k^2-m^2}\,\frac{\gamma^+}{2P^+}\,\frac{i(\not{k}+m)}{k^2-m^2}\,(ig_s)\,u(P,S)2\pi\,\delta\big[(P-k)^2-\lambda^2\big] \\ &= \frac{g_s^2}{2}\int\frac{dk^-}{2\pi}\,\frac{\bar{u}(P,S)(\not{k}+m)\gamma^+(\not{k}+m)u(P,S)}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{k^+=xP^+} \end{aligned} \tag{29}$$

Averaging over the nucleon spin

$$f(x,k_\perp^2)=\frac{1}{2}\sum_S\frac{g_s^2}{2}\int\frac{dk^-}{2\pi}\,\frac{\bar{u}(P,S)(\not{k}+m)\gamma^+(\not{k}+m)u(P,S)}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{k^+=xP^+} \tag{30}$$

$$\begin{aligned} \sum_S\bar{u}(P,S)(\not{k}+m)\gamma^+(\not{k}+m)u(P,S) &= Tr\Big[(\not{k}+m)\gamma^+(\not{k}+m)(\not{P}+M)\Big] \\ &= 4k^+(k.P)-2P^+k^2+4mMk^++2m^2P^+\Big|_{x=k^+/P^+} \\ &= 2P^+[k_T^2+(m+xM)^2] \end{aligned} \tag{31}$$

Using this in the numerator

$$f(x,k_\perp^2)=\frac{g_s^2P^+}{2}\int\frac{dk^-}{2\pi}\,\frac{[k_T^2+(m+xM)^2]}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{x=k^+/P^+} \tag{32}$$

Taking the k^- integral using the delta-function

$$\begin{aligned} f(x,k_\perp^2) &= \frac{g_s^2}{4}\int\frac{dk^-}{2\pi}\,\frac{[k_T^2+(m+xM)^2]}{(2xP^+k^--k_T^2-m^2)^2}\,\frac{1}{(1-x)}\delta\Big(k^--\frac{M^2}{2P^+}+\frac{(k_T^2+\lambda^2)}{2P^+(1-x)}\Big) \\ &= \frac{g_s^2}{4}\,\frac{1}{2\pi}\,\frac{(1-x)[k_T^2+(m+xM)^2]}{(k_T^2+u)^2} \end{aligned} \tag{33}$$

$u=-x(1-x)M^2+(1-x)m^2+x\lambda^2$ The PDF is

$$f(x,\mu) \quad = \quad \frac{g_s^2}{8\pi}(1-x)\mu^{2\epsilon}\int\frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\,\frac{[k_T^2+(m+xM)^2]}{(k_T^2+u)^2} \tag{34}$$

The transverse integrals:

$$\mu^{2\epsilon}\int \frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{k_T^2}{(k_T^2+u)^2} = \frac{\mu^{2\epsilon}}{(2\pi)^{2-2\epsilon}}\frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\int_0^\infty dk_T^2\frac{(k_T^2)^{1-\epsilon}}{(k_T^2+u)^2} \tag{35}$$
$$= \frac{\mu^{2\epsilon}}{(2\pi)^{2-2\epsilon}}\frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\frac{\Gamma(2-\epsilon)\Gamma(\epsilon)}{\Gamma(2)u^\epsilon}$$
$$= \frac{1}{4\pi}\Big(\frac{1}{\epsilon}-\gamma_E+\ln 4\pi+\ln\frac{\mu^2}{u}-1\Big)+\mathcal{O}(e)$$
$$\to \frac{1}{4\pi}\Big(\mathcal{D}_{UV}+\ln\frac{\mu^2}{u}-1\Big)$$

$$\mu^{2\epsilon}\int \frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{1}{(k_T^2+u)^2} = \frac{\mu^{2\epsilon}}{(2\pi)^{2-2\epsilon}}\frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}\int_0^\infty dk_T^2\frac{(k_T^2)^{1-\epsilon}}{(k_T^2+u)^2} \tag{36}$$
$$= \frac{\mu^{2\epsilon}}{(2\pi)^{2-2\epsilon}}\frac{\pi^{1-\epsilon}}{\Gamma(-\epsilon)}\frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2)u^{1+\epsilon}}$$
$$= \frac{1}{4\pi}\frac{1}{u}+\mathcal{O}(e)$$

Using these results we get the PDF as

$$f(x,\mu) = \frac{g^2}{32\pi^2}(1-x)\Big(\mathcal{D}_{UV}+\ln\frac{\mu^2}{u}-1+\frac{(m+xM)^2}{u}\Big) \tag{37}$$

3.2 pPDF

$$f(x,z_T) = \frac{g_s^2}{8\pi}(1-x)\int\frac{d^2k_T}{(2\pi)^2}e^{ik_T.z_T}\frac{[k_T^2+(m+xM)^2]}{(k_T^2+u)^2} \tag{38}$$

This time the transverse integrals

$$\int\frac{d^2k_T}{(2\pi)^2}\frac{e^{ik_T.z_T}}{(k_T^2+u)^2} = \frac{1}{4\pi}\frac{|z|}{\sqrt{u}}K_1\sqrt{u\,z_T^2} \tag{39}$$

$$\int\frac{d^2k_T}{(2\pi)^2}e^{ik_T.z_T}\frac{k_T^2}{(k_T^2+u)^2} = \frac{1}{2\pi}K_0\sqrt{u\,z_T^2}-\frac{u}{4\pi}\frac{|z|}{\sqrt{u}}K_1\sqrt{u\,z_T^2} \tag{40}$$

$$f(x,z_T)=\frac{g_s^2}{32\pi^2}(1-x)\Bigg(2K_0\sqrt{u\,z_T^2}+\Big[(m+xM)^2-u\Big]\frac{|z|}{\sqrt{u}}K_1\sqrt{u\,z_T^2}\Bigg) \tag{41}$$

Using Small-argument expansion of the modified Bessel function $K_0(y)$ and $K_1(y)$

$$K_0(y)=-\gamma_E-\ln(\frac{y}{2})+O(y^2), \qquad y>0. \tag{42}$$

$$K_1(y)=\frac{1}{y}+\frac{y}{2}\left(\ln\frac{y}{2}+\gamma_E-\frac{1}{2}\right)+O\big(y^3\ln y\big)\,.$$

$$f(x,z_T)_{Truncated}=\frac{g_s^2}{32\pi^2}(1-x)\Bigg(-2\gamma_E-\ln\frac{uz_T^2}{4}+\frac{(m+xM)^2-u}{u}\Bigg) \tag{43}$$

3.3 Zeroth order matching in Scalar Diquark Model

$$f(x,\mu)=\frac{g^2}{32\pi^2}(1-x)\Big(\ln\frac{\mu^2}{u}-1+\frac{(m+xM)^2}{u}\Big) \tag{44}$$

$$f(x,z_T)_{Truncated}=\frac{g^2}{32\pi^2}(1-x)\Bigg(-2\gamma_E-\ln\frac{uz_T}{4}+\frac{(m+xM)^2-u}{u}\Bigg) \tag{45}$$

$$f(x,z_T)_{Truncated}-f(x,\mu)=\frac{g^2}{32\pi^2}(1-x)\Big(-2\gamma_E-\ln(\mu^2z_T^2)+\ln 4\Big) \tag{46}$$

4 Axial Diquark Model: f_1

The TMD is

$$f(x,k_\perp^2) = \int \frac{dk^+dk^-}{(2\pi)^2} P^+\delta\Big(k^+ - xP^+\Big)\, \bar{u}(P,S)\,(ig_a\gamma^\mu\gamma_5)\,\frac{i(\not{k}+m)}{k^2-m^2}\,\frac{\gamma^+}{2P^+}\,\frac{i(\not{k}+m)}{k^2-m^2}\,(ig_a\gamma^\nu\gamma_5)\,u(P,S)d^{\mu\nu}2\pi\,\delta\big[(P-k)^2-\lambda^2\big] \quad (47)$$

Light-cone PDF polarization tensor ($n^2=0$)

$$d^{\mu\nu}(P-k,n) = -g^{\mu\nu} + \frac{n^\mu(P-k)^\nu + n^\nu(P-k)^\mu}{P^+-k^+} - \frac{(P-k)^2\,n^\mu n^\nu}{(P^+-k^+)^2} \quad (48)$$

$$f(x,k_\perp^2) = \frac{g_a^2}{2}\int\frac{dk^-}{2\pi}\,\frac{\bar{u}(P,S)\gamma^\mu\gamma_5(\not{k}+m)\gamma^+(\not{k}+m)\gamma^\nu\gamma_5 u(P,S)d^{\mu\nu}}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{k^+=xP^+} \quad (49)$$

Averaging over the nucleon spin

$$f(x,k_\perp^2) = \frac{1}{2}\sum_S\frac{g_a^2}{2}\int\frac{dk^-}{2\pi}\,\frac{\bar{u}(P,S)\gamma^\mu\gamma_5(\not{k}+m)\gamma^+(\not{k}+m)\gamma^\nu\gamma_5 u(P,S)d^{\mu\nu}}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{k^+=xP^+} \quad (50)$$

$$\sum_S \bar{u}(P,S)\gamma^\mu\gamma_5(\not{k}+m)\gamma^+(\not{k}+m)\gamma^\nu\gamma_5 u(P,S)d^{\mu\nu} = Tr\Big[\gamma^\mu\gamma_5(\not{k}+m)\gamma^+(\not{k}+m)\gamma^\nu\gamma_5(\not{P}+M)\Big]d_{\mu\nu} \quad (51)$$

$$\begin{aligned}
&= \frac{8P^+}{(1-x)^2}\Big[(1+x^2)(2x(k.P)-k^2) + (1-x)^2m^2 + 2x(1-x)^2mM - 2M^2x^3\Big] \\
&= \frac{8P^+}{(1-x)^2}\Big[(1+x^2)(x^2M^2+k_T^2) + (1-x)^2m^2 + 2x(1-x)^2mM - 2M^2x^3\Big] \\
&= \frac{8P^+}{(1-x)^2}\Big[(1+x^2)k_T^2 + (1-x)^2(m+xM)^2\Big] \\
&= 8P^+\Big[\frac{(1+x^2)}{(1-x)^2}k_T^2 + (m+xM)^2\Big]
\end{aligned}$$

Plugging this in the numerator

$$f(x,k_\perp^2) = g_a^2\;2P^+\int\frac{dk^-}{2\pi}\,\frac{\Big[\frac{(1+x^2)}{(1-x)^2}k_T^2 + (m+xM)^2\Big]}{(k^2-m^2)^2}\delta\big[(P-k)^2-\lambda^2\big]\Big|_{k^+=xP^+} \quad (52)$$

$$f(x,k_\perp^2) \;=\; g_a^2\;2P^+\int\frac{dk^-}{2\pi}\frac{\Big[\frac{(1+x^2)}{(1-x)^2}k_T^2+(m+xM)^2\Big]}{(2xP^+k^- - k_T^2 - m^2)^2}\frac{1}{2P^+(1-x)}\delta\Big(k^- - \frac{M^2}{2P^+}+\frac{(k_T^2+\lambda^2)}{2P^+(1-x)}\Big) \tag{53}$$
$$= \frac{g_a^2}{2\pi}\frac{(1-x)\Big[\frac{(1+x^2)}{(1-x)^2}k_T^2+(m+xM)^2\Big]}{(k_T^2+u)^2}$$

$u=-x(1-x)M^2+(1-x)m^2+x\lambda^2$

Using dim reg to take the transverse integrals

$$f(x,\mu) \;=\; \frac{g_a^2}{2\pi}\frac{(1+x^2)}{(1-x)}\,\mu^{2\epsilon}\int\frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{k_T^2}{(k_T^2+u)^2}+\frac{g_a^2}{2\pi}\,(1-x)(m+xM)^2\mu^{2\epsilon}\int\frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{1}{(k_T^2+u)^2} \tag{54}$$

$$\mu^{2\epsilon}\int\frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{k_T^2}{(k_T^2+u)^2}=\frac{1}{4\pi}\Big(\frac{1}{\epsilon}-\gamma_E+\ln 4\pi+\ln\frac{\mu^2}{u}-1\Big)+\mathcal{O}(e)\to\frac{1}{4\pi}\Big(\mathcal{D}_{UV}+\ln\frac{\mu^2}{u}-1\Big) \tag{55}$$

$$\mu^{2\epsilon}\int\frac{d^{2-2\epsilon}k_T}{(2\pi)^{2-2\epsilon}}\frac{1}{(k_T^2+u)^2} \;=\; \frac{1}{4\pi}\frac{1}{u}+\mathcal{O}(e) \tag{56}$$

$$f(x,\mu)=\frac{g_a^2}{8\pi^2}\frac{(1+x^2)}{(1-x)}\Big(\mathcal{D}_{UV}+\ln\frac{\mu^2}{u}-1\Big)+\frac{g_a^2}{8\pi^2}\,(1-x)(m+xM)^2\frac{1}{u} \tag{57}$$

4.1 pPDFs

$$f(x,z_T)=\frac{g_a^2}{2\pi}\frac{(1+x^2)}{(1-x)}\int\frac{d^2k_T}{(2\pi)^2}e^{ik_T.z_T}\frac{k_T^2}{(k_T^2+u)^2}+\frac{g_a^2}{2\pi}\,(1-x)(m+xM)^2\int\frac{d^2k_T}{(2\pi)^2}e^{ik_T.z_T}\frac{1}{(k_T^2+u)^2} \tag{58}$$

$$\int\frac{d^2k_T}{(2\pi)^2}\frac{e^{ik_T.z_T}}{(k_T^2+u)^2}=\frac{1}{4\pi}\frac{|z|}{\sqrt{u}}K_1\sqrt{u\,z_T^2} \tag{59}$$

$$\int \frac{d^2k_T}{(2\pi)^2} e^{ik_T.z_T} \frac{k_T^2}{(k_T^2+u)^2} = \frac{1}{2\pi} K_0 \sqrt{u\, z_T^2} - \frac{u}{4\pi} \frac{|z|}{\sqrt{u}} K_1 \sqrt{u\, z_T^2} \tag{60}$$

$$f(x, z_T) \;\; = \;\; \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(2K_0 \sqrt{u\, z_T^2} - \frac{u|z|}{\sqrt{u}} K_1 \sqrt{u\, z_T^2} \Big) + \frac{g_a^2}{8\pi^2} (1-x)(m+xM)^2 \frac{|z_T|}{\sqrt{u}} K_1 \sqrt{u\, z_T^2} \tag{61}$$

$$f(x, z_T) \;\; = \;\; \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(2(-\gamma_E - \ln \frac{\sqrt{u z_T^2}}{2}) - \frac{u|z_T|}{\sqrt{u}} \frac{1}{\sqrt{u\, z_T^2}} \Big) + \frac{g_a^2}{8\pi^2} (1-x)(m+xM)^2 \frac{|z_T|}{\sqrt{u}} \frac{1}{\sqrt{u\, z_T^2}} \tag{62}$$

$$f(x, \mu)_{Truncated} = \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(-2\gamma_E - \ln(u z_T^2) + \ln 4 - 1 \Big) + \frac{g_a^2}{8\pi^2} (1-x)(m+xM)^2 \frac{1}{u} \tag{63}$$

4.2 The matching

$$f(x, \mu) = \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(\mathcal{D}_{UV} + \ln \frac{\mu^2}{u} - 1 \Big) + \frac{g_a^2}{8\pi^2} (1-x)(m+xM)^2 \frac{1}{u} \tag{64}$$

$$f(x, \mu)_{Truncated} = \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(-2\gamma_E - \ln(u z_T^2) + \ln 4 - 1 \Big) + \frac{g_a^2}{8\pi^2} (1-x)(m+xM)^2 \frac{1}{u} \tag{65}$$

$$f(x, \mu)_{Truncated} - f(x, \mu) = \frac{g_a^2}{8\pi^2} \frac{(1+x^2)}{(1-x)} \Big(-2\gamma_E - \ln(\mu^2 z_T^2) + \ln 4 \Big) \tag{66}$$